



GATE SMASHERS

GATE2024

Engineering Mathematics

Topic: Probability and Statistics

Subtopic: Combination Techniques

LECTURE - 7

EXPLANATION

Example 1 - A team of 6 members is to be formed by selecting out of 5 men and 6 women. In how many ways the committee can be formed if it contains

(i) Exact 2 men. (ii) at least one man?



Example 2 - Out of 5 consonants and 4 vowels, how many words each containing 3 consonants and 2 vowels can be formed?

Example 3 - A boy has 3 library tickets and 8 books of his interest in the library. Of these 8 books, he does not want to borrow Maths part 2, unless Maths part 1 is also borrowed. In how many ways he chooses three books?

Example 4 - How many diagonals can be drawn by joining vertices of hexagon.

Example 5 - How many straight lines can be formed by joining 12 points, 7 of which are on the same line.

Example 6 - In a room every body shakes hands with everybody else. Total no. of handshakes are 6. Find total number of persons in the room.

NOTES PREPARED BY: ANKIT ARORA